

# Polar Coordinates and Conversions



## Converting points

- 1 Convert each polar point  $(r, \theta)$  to rectangular coordinates (exact values):

**a**  $(4, \frac{\pi}{3})$

**b**  $(2, \frac{3\pi}{4})$

- 2 Convert each rectangular point to polar coordinates with  $r \geq 0$  and  $0 \leq \theta < 2\pi$ :

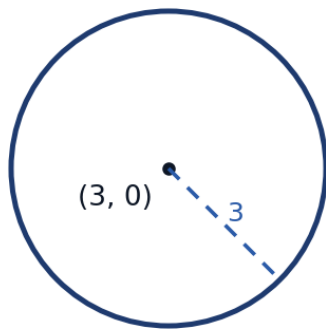
**a**  $(-3, 3)$

**b**  $(0, -5)$

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## Converting equations

- 3 Convert the polar equation  $r = 6\cos\theta$  to rectangular form and identify the curve.

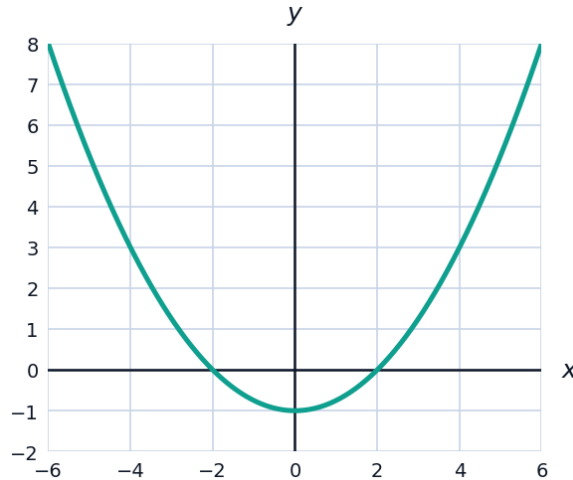


- 4 Convert each rectangular equation to polar form:

**a**  $x^2 + y^2 = 16$

**b**  $y = x$

- 5 Convert the polar equation  $r = \frac{2}{1 - \sin \theta}$  to rectangular form.

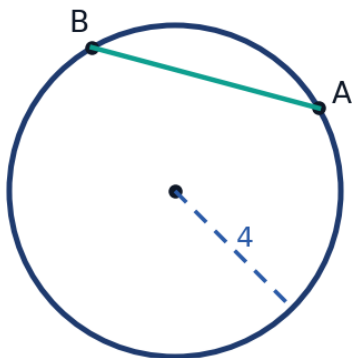


### Multiple representations

- 6 Give two other pairs  $(r, \theta)$  that represent the same point as  $(3, \frac{\pi}{4})$ : one with  $r > 0$  and  $\theta$  outside  $[0, 2\pi)$ , and one with  $r < 0$ .
- 7 Plot-reasoning: explain why the polar point  $(-2, \frac{\pi}{2})$  corresponds to the same location as  $(2, \frac{3\pi}{2})$ .

### Distance between polar points

- 8 Convert  $(4, \frac{\pi}{6})$  and  $(4, \frac{2\pi}{3})$  to rectangular coordinates, then find the distance between them, to two decimal places.



## Further conversion practice

9 Convert each polar point to rectangular coordinates (exact values):

a  $(6, \pi)$

b  $(5, \frac{5\pi}{6})$

10 Convert the polar equation  $r^2 = 4\sin 2\theta$  to note its symmetry, and identify it as a standard curve type.

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## Why polar coordinates are not unique

11 Explain why, unlike rectangular coordinates, a single point can be represented by infinitely many polar coordinate pairs.

12 The pole (origin) is a special case in polar coordinates. Explain why.

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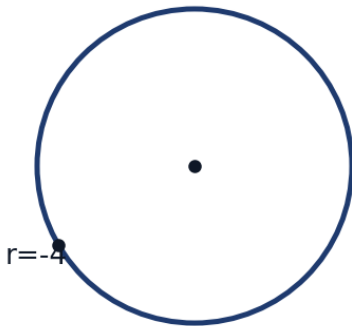
## Converting more complex equations

13 Convert the rectangular equation  $x^2 - y^2 = 4$  (a hyperbola) to polar form.

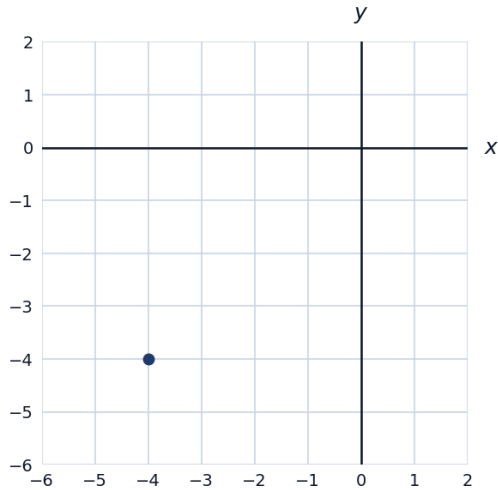
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## More point conversions with negative r

14 Convert the polar point  $(-4, \frac{\pi}{6})$  (negative  $r$ ) to rectangular coordinates.



- 15 Convert the rectangular point  $(-4, -4)$  to polar coordinates with  $r > 0$  and  $0 \leq \theta < 2\pi$ .

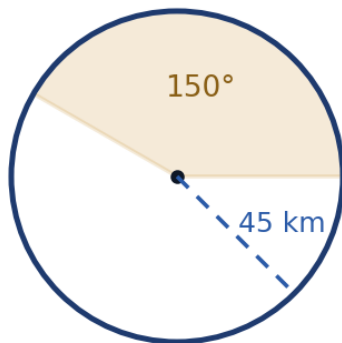


### Converting a spiral equation

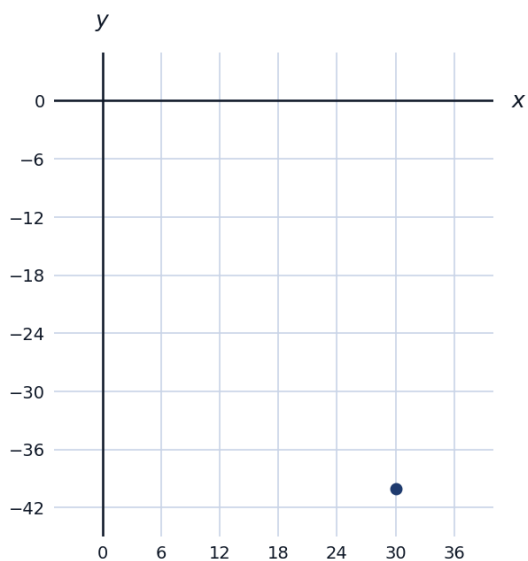
- 16 The Archimedean spiral  $r = \theta$  has no simple rectangular equivalent, but explain qualitatively why it spirals outward as  $\theta$  increases.

### Converting a real-world radar equation

- 17 A radar station models detected objects using polar coordinates  $(r, \theta)$  where  $r$  is distance in km and  $\theta$  is bearing angle. An aircraft is detected at  $(r, \theta) = (45, \frac{5\pi}{6})$ . Convert this to rectangular coordinates (kilometers east and north of the station), to two decimal places.



- 18** A second aircraft is detected at rectangular coordinates  $(30, -40)$  relative to the same radar station. Convert to polar coordinates  $(r, \theta)$  with  $r > 0$  and  $0 \leq \theta < 2\pi$ , to two decimal places.



- 19** Convert the rectangular equation of a vertical line,  $x = 5$ , to polar form, and explain why the result is not simply a constant  $r$ .
- 20** Convert the polar equation  $r = 6$  to rectangular form, and explain in one sentence why this equation contains no  $\theta$  term.

